

Topic 1: Formulae, Equations and Amount of Substance

Application of ideas from this topic will be applied to all other units.

Students will be assessed on their ability to:

1.1	know the terms 'atom', 'element', 'ion', 'molecule', 'compound', 'empirical formula' and 'molecular formula'
1.2	know that the mole (mol) is the unit for the amount of a substance and be able to perform calculations using the Avogadro constant L ($6.02 \times 10^{23} \text{ mol}^{-1}$)
1.3	write balanced full and ionic equations, including state symbols, for chemical reactions
1.4	understand the terms: - i 'relative atomic mass' based on the ^{12}C scale - ii 'relative molecular mass' and 'relative formula mass', including calculating these values from relative atomic masses <i>The term 'relative formula mass' should be used for compounds with giant structures.</i> - iii 'molar mass' as the mass per mole of a substance in g mol^{-1} - iv parts per million (ppm), including gases in the atmosphere
1.5	calculate the concentration of a solution in mol dm^{-3} and g dm^{-3} <i>Titration calculations are not required at this stage.</i>
1.6	be able to use experimental data to calculate empirical and molecular formulae
1.7	be able to use chemical equations to calculate reacting masses and vice versa, using the concepts of amount of substance and molar mass
1.8	be able to use chemical equations to calculate volumes of gases and vice versa, using: - i the concepts of amount of substance - ii the molar volume of gases - iii the expression $pV = nRT$ for gases and volatile liquids
1.9	be able to calculate percentage yields and percentage atom economies (by mass) in laboratory and industrial processes, using chemical equations and experimental results Atom economy = $\frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all products}} \times 100\%$
1.10	be able to determine a formula or confirm an equation by experiment, including evaluation of the data
1.11	CORE PRACTICAL 1 Measurement of the molar volume of a gas.
1.12	be able to relate ionic and full equations, with state symbols, to observations from simple test-tube experiments, to include: - i displacement reactions - ii typical reactions of acids - iii precipitation reactions

Further suggested practicals:

- i preparation of a salt and calculating the percentage yield of product, including the preparation of a double salt, such as ammonium iron(II) sulfate from iron, ammonia and sulfuric acid
- ii determine a chemical formula by experiment, such as the formula of copper(II) oxide by reduction
- iii determine a chemical equation by experiment, such as the reaction between lithium and water, or the reaction between magnesium and an acid
- iv carry out and interpret the results of simple test-tube reactions, as outlined in 1.12